

Blaine Heffron, PhD

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scholar.google.com/citations?user=8d-y4zoAAAAJ

Physics AI training profile

Physics PhD with a research background in neutrino physics, detector analysis, statistical inference, simulation, and machine learning. I have written and reviewed technical explanations, built reproducible analysis software, and checked noisy detector or model output against physical constraints. I am a good match for AI training work that needs hard physics problems, careful review of model reasoning, and clear explanations of what went wrong.

AI model training relevance

- Create and validate advanced physics tasks in particle and nuclear physics, detector physics, statistical data analysis, classical mechanics, electromagnetism, quantum mechanics, and computational physics.
- Trained to evaluate derivations, units, boundary conditions, approximations, numerical results, statistical claims, and hidden assumptions in complex technical answers.
- Built ML and signal-processing workflows for PROSPECT, including CNN and graph neural network approaches for event reconstruction and classification under noisy detector conditions.
- Write prompts, reference solutions, and grading notes; follow detailed rubrics while separating correct physics from answers that only sound right.

Education

PhD, Physics , University of Tennessee, Knoxville	GPA: 3.98/4.0	2023
MS, Physics , University of Tennessee, Knoxville	GPA: 4.0/4.0	2017
BS, Physics , University of North Carolina at Chapel Hill	GPA: 3.7/4.0	2012

Research experience

Graduate Research Assistant <i>University of Tennessee, Department of Physics</i>	Aug 2019 – Jan 2023 <i>Knoxville, TN</i>
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- Developed analysis software in Python and C++ for the PROSPECT reactor neutrino experiment at Oak Ridge National Laboratory.
- Applied machine learning to event reconstruction and background rejection, including graph neural networks for particle identification and sparse CNN methods for position and energy reconstruction.
- Improved single-ended detector reconstruction in work later published by the collaboration, increasing usable statistics for PROSPECT analysis.
- Planned and executed HFIR radiation field measurements with HPGe instrumentation; analyzed large datasets on reactor gamma backgrounds and detector constraints.
- Mentored undergraduate and graduate researchers on physics analysis, software workflows, debugging, and technical communication.

Graduate Research Assistant <i>University of Tennessee, Department of Physics</i>	Aug 2014 – Dec 2016 <i>Knoxville, TN</i>
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- Built simulation and analysis tooling for reactor and cosmogenic background studies supporting the PROSPECT detector program.
- Produced documented research software and technical analysis for detector characterization, shielding studies, and experimental planning.

Research Assistant <i>UNC Chapel Hill, Environmental Sciences and Engineering</i>	Nov 2012 – Aug 2014 <i>Chapel Hill, NC</i>
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- Built Python modeling tools for sunlight flux and smog chamber simulation, including nonlinear differential equation models for aerosol formation and dynamics.

Selected publications and research output

- **Machine learning for single-ended event reconstruction in PROSPECT experiment.** M. Andriamirado et al. (PROSPECT Collaboration), *JINST* 20, P08006 (2025).
- **Final Measurement of the U-235 Antineutrino Energy Spectrum with the PROSPECT-I Detector at HFIR.** M. Andriamirado et al. (PROSPECT Collaboration), *Phys. Rev. Lett.* 131, 021802 (2023).
- **Joint Measurement of the U-235 Antineutrino Spectrum by PROSPECT and STEREO.** H. Almazan, M. Andriamirado et al., *Phys. Rev. Lett.* 128, 081802 (2022).
- Public research materials include PROSPECT ML summary slides and open-source WaveformML code linked from blaineheffron.com.

Software, AI, and technical experience

Independent Consultant / Software Developer

Feb 2023 – Present

Self-Employed

Charleston, SC

- Build AI automation, workflow tools, internal dashboards, data pipelines, and research-adjacent software in Python, TypeScript, Rust, C++, and SQL.
- Use LLMs in applied systems where generated output has to be checked, routed, documented, or reviewed before anyone trusts it.
- Developed paper-search and symbolic-math MCP tools for scientific paper search, local indexing, symbolic mathematics, and tensor algebra workflows.
- Turn messy technical requirements into working systems with clear documentation, validation paths, and handoff notes.

Software Engineer / Smart Contract Tooling Contractor

2024 – 2025

AhaLabs / Stellar Ecosystem

Remote

- Built Rust and TypeScript developer tooling for Stellar smart contract development, including generated bindings, deployment tooling, and build/watch workflows.
- Won a \$60k Stellar Community Fund grant for soroban-sdk-tools, an open-source Rust crate reducing contract boilerplate around storage, errors, imports, and auth testing.

Technical Support Engineer

Mar 2018 – Aug 2019

CData Software

Remote

- Supported enterprise Java and C# data connectivity products across databases, APIs, and application environments.
- Reproduced and resolved complex technical issues, documented fixes, and built tooling to generate integrations from OpenAPI specifications.

Technical skills

Physics: Particle physics, nuclear physics, neutrino physics, detector analysis, radiation backgrounds, HPGe measurements, signal processing, simulation, statistical analysis, uncertainty reasoning

AI/ML: PyTorch, PyTorch Geometric, CNNs, graph neural networks, classification, regression, feature engineering, model evaluation, data validation

Programming: Python, C++, Rust, TypeScript, JavaScript, SQL, Java, C#, Linux, Git

Communication: Research writing, technical documentation, code review, mentoring, rubric-based evaluation, precise written feedback

Honors

Extraordinary Professional Promise, UT Knoxville

2016

Stelson Fellowship, UT Knoxville

2016

Chancellor's Scholarship, UT Knoxville

2014–2016

Robert C. Byrd Scholarship, UNC Chapel Hill

2008–2012